IGC: Training Architecture and Gradient Flow

(draft figure — merge into the main paper)

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1 Training decomposition

IGC is trained as *three separate supervised/RL "islands," never end-to-end.* This is a deliberate design choice (and is what the current implementation enforces with hard stop-gradients), not an accident of the code:

- **State Encoder (M1)** is the only module trained end-to-end *within itself*: a causal-LM next-token loss backpropagates through the entire backbone under a single optimizer. It learns a representation of a Redfish observation (a GET/HEAD response).
- **State Pooler (M2)** learns a compact bottleneck by reconstructing the encoder's last hidden state; the backbone runs under `no_grad`, so the reconstruction gradient reaches only the pooler.
- **Policy (M6)** is trained by temporal-difference RL (with hindsight experience replay) and updates *only* the action-value network. The encoder is **frozen** during RL: the observation embedding is detached at the environment boundary, so no RL gradient ever flows back into the representation. This keeps the (expensive, language-pretrained) representation stable while the policy explores.

Reward and action edges are *non-differentiable* (they carry data into the TD target, not gradient). The evaluator (M4) and world-model (M5) are part of the planned design; the accepted pointer-style policy head is implemented but not yet wired into the training loop. Figure 1 makes the gradient boundaries explicit.

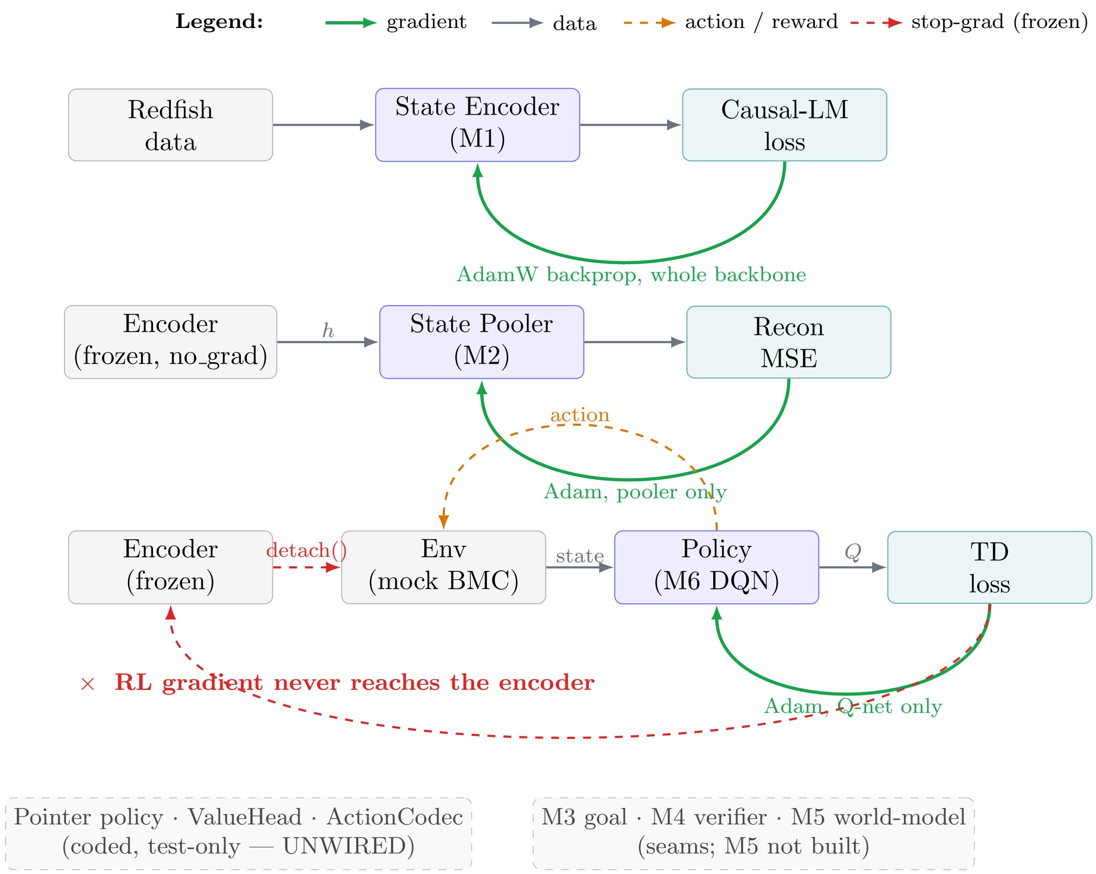

Figure 1: IGC gradient flow. Three training islands with hard stop-gradients between them. **M1** (state encoder) is trained by a causal-LM loss through the whole backbone. **M2** (state pooler) reconstructs the encoder's hidden state with the backbone frozen. **M6** (policy) is trained by TD/HER and updates only the action-value network; the encoder is frozen during RL (the observation embedding is detached at the environment boundary), so *no RL gradient reaches the representation.* Action and reward edges are non-differentiable. The pointer policy, value head, and the M3/M4/M5 components are part of the design but not yet in the live training loop.